Yimo Kong

Software Engineering | C++ Graphics & Real-Time Systems

Singapore · kongyimoo@gmail.com · +65 8531 0536 · github.com/kongyimo1224 · linkedin.com/in/yimokong1224

EDUCATION

Singapore Institute of Technology × DigiPen Institute of Technology 2024 – Expected 2028
BSc (Hons) Computer Science in Real-Time Interactive Simulation · Overseas Immersion, DigiPen Europe-Bilbao (2026)

Nanyang Polytechnic 2021 – 2024
Diploma in Infocomm & Media Engineering · Director's List

EXPERIENCE

Tech Lead — Upsight × DigiPen Industry Project Sep 2026 – Expected March 2027

- Lead an 8-person engineering team delivering ChargedUp, an EV journey-planning platform for an industry client across six planned milestones.
- Engineered a Babylon.js 3D city renderer integrated with MapLibre GL; generated procedural meshes from vector tiles using Web Workers, visibility culling and adaptive LOD.
- Built TypeScript charger-ranking and vehicle-aware stop recommendations; established 200+ automated tests, GitHub Actions CI and Docker/Jenkins deployment workflows.

Software Development & Data Analytics Intern — GERMS Digital Sep 2023 – Feb 2024

- Developed SYFER, a dashboard aggregating Google Trends, Meta API and RSS data with daily refreshes, adopted company-wide; maintained UOB campaign websites.

SELECTED PROJECTS

Bloody Good Curry — custom 2D game engine & shipped game Sep 2025 – Apr 2026
C++17, OpenGL/GLSL, ImGui, Emscripten (WebAssembly + WebGL 2), FMOD, CMake | Play: bloodygoodcurry.netlify.app

- Co-built a ~53K-line C++ engine and editor from scratch in a 7-person team: component-based objects, JSON-serialized levels and prefabs, pooled allocators, 2D physics with grid broad phase and AI decision trees; one codebase ships to Windows and the browser.
- Owned the rendering pipeline: render system with game and editor viewports, orthographic camera, screen-to-world picking, procedural glow lighting and combat VFX; GPU-instanced sprite batching cut draw calls **38% (250 → 156 per frame)** in the densest level.
- Built the engine core loop, a per-frame CPU profiler and much of the ImGui editor: transform gizmo, sprite-sheet animation editor, hierarchy, inspector, JSON data editor, layer and prefab-spawn panels.

Flatland Bound — networked co-op puzzle platformer (solo) Sep 2026 – Oct 2026
C11, OpenGL 3.3, GLFW, Cyclone DDS (DDS Security), Linux/WSL

- Two players see one 3D voxel world from perpendicular sides and turn it to any angle; built any-angle projected collision, deterministic physics that runs mid-turn, and a renderer with shadows, HDR bloom and ACES.
- Implemented client-side prediction with server reconciliation over a sequence-locked per-tick input stream: input-to-screen latency stays at one frame **(17 ms) at any emulated ping vs. 135–318 ms** without it; only 4 of 928 snapshots needed correction under 10% input loss.
- NTP-style clock sync aligns turn starts within 2–5 ms at 100 ms emulated latency; an event-driven, lock-free server (DDS waitsets, SPSC queues) cut idle CPU **199% → 1.7%**.

Real-Time Rendering & Spatial Acceleration 2025 – 2026
C++, OpenGL, GLSL

- Implemented 11 shader/rendering techniques including deferred shading, HDR environment mapping, toon shading, portal rendering and ray tracing.
- Built BVH and SAH-based KD-tree ray acceleration with closest-hit traversal; developed a CPU rasterization pipeline with frustum culling, clip-space polygon clipping, depth buffering, texturing and per-pixel lighting.

四水归堂 Four Waters Return to the Hall — Tsinghua School of Architecture × miHoYo Competition 2026
Unreal Engine 5, C++, Python, Blender

- First-person puzzle game in a Huizhou courtyard house, built in a cross-disciplinary team of three.
- ~8,700 lines of UE5 gameplay C++: interactions, four water puzzles, and an era switch using a SceneCapture reflection and wipe transition; Lumen, Nanite and Python-scripted level building.

TECHNICAL SKILLS

Languages	C++, C, C#, Python, TypeScript/JavaScript, SQL, GLSL
Graphics & Engines	OpenGL, GLFW/GLM, ImGui, Unity, Unreal Engine, Babylon.js, Blender
Developer Tools	Git, Linux/WSL, Cyclone DDS, CMake, Visual Studio, Docker, GitHub Actions, Jenkins